



Design Pulse-Width Discriminator for Positive-Going Pulse

05366023 Electronic Circuit

วงจรอิเล็กทรอนิกส์

จัดทำโดย

รหัสนักศึกษา : 67050066

ชื่อ - นามสกุล : นางสาวจิตาภา จินดาประเสริฐ

รหัสนักศึกษา : 67050128

ชื่อ - นามสกุล : นางสาวจิตติมา แนวพระยา

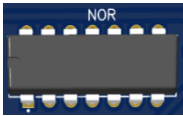
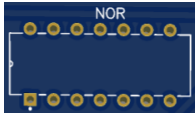
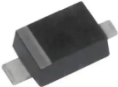
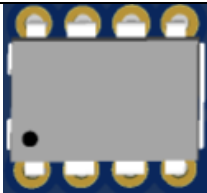
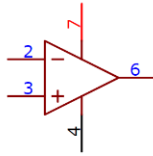
รหัสนักศึกษา : 67050326

ชื่อ - นามสกุล : นายปัญญา ไตรพรหม

นักศึกษา ภาควิชาฟิสิกส์อุตสาหกรรม
สาขา ฟิสิกส์อุตสาหกรรมและวิศวกรรมศาสตร์ไอโอทีและสารสนเทศ
คณะวิทยาศาสตร์

สถาบันเทคโนโลยีพระจอมเกล้าเจ้าคุณทหารลาดกระบัง

ชื่ออุปกรณ์และสัญลักษณ์ของอุปกรณ์ที่ใช้แบบ Through-Hole Technology (THT)

ชื่ออุปกรณ์	ภาพอุปกรณ์	3D PCB	สัญลักษณ์	สัญลักษณ์บน PCB
Nor-gate_7402				
1N4148WT_SMD				
Zener Diode	1N4566 			
	1N752A 			
LM741_Op-Amp				
Resistor				
Capacitor_SMD				

หมายเหตุ

1N4148WT_SMD : D₁, D₂, D₃, D₄, D₅, D₆

1N4566 : Z₁, Z₂, Z₃, Z₅, Z₆

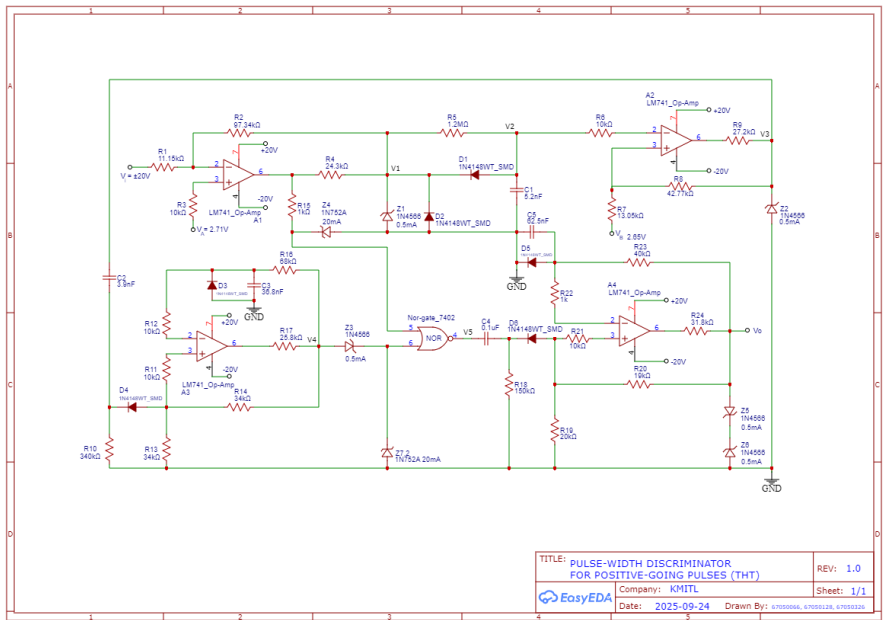
1N752A : Z₄, Z₇

LM741_Op-Amp : A₁, A₂, A₃, A₅

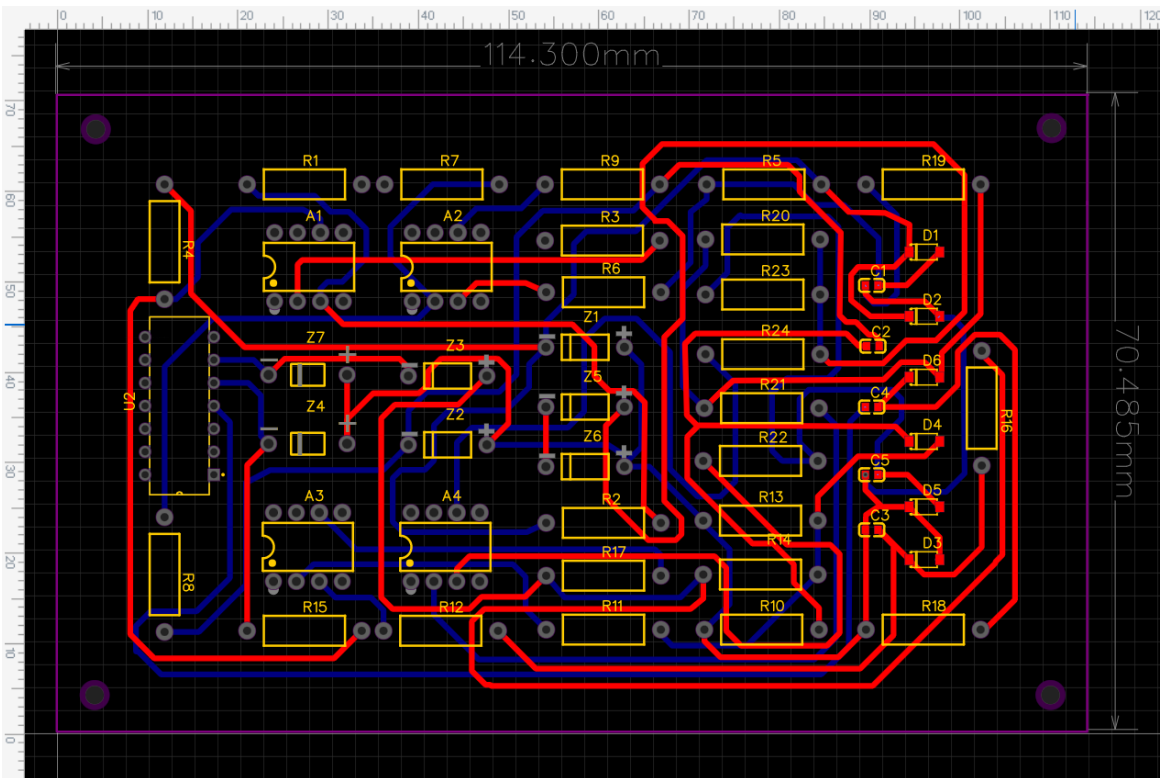
Through-Hole Technology (THT) : แบบเจาะรู

Schematic Diagram of PULSE-WIDTH DISCRIMINATOR FOR POSITIVE-GOING PULSES

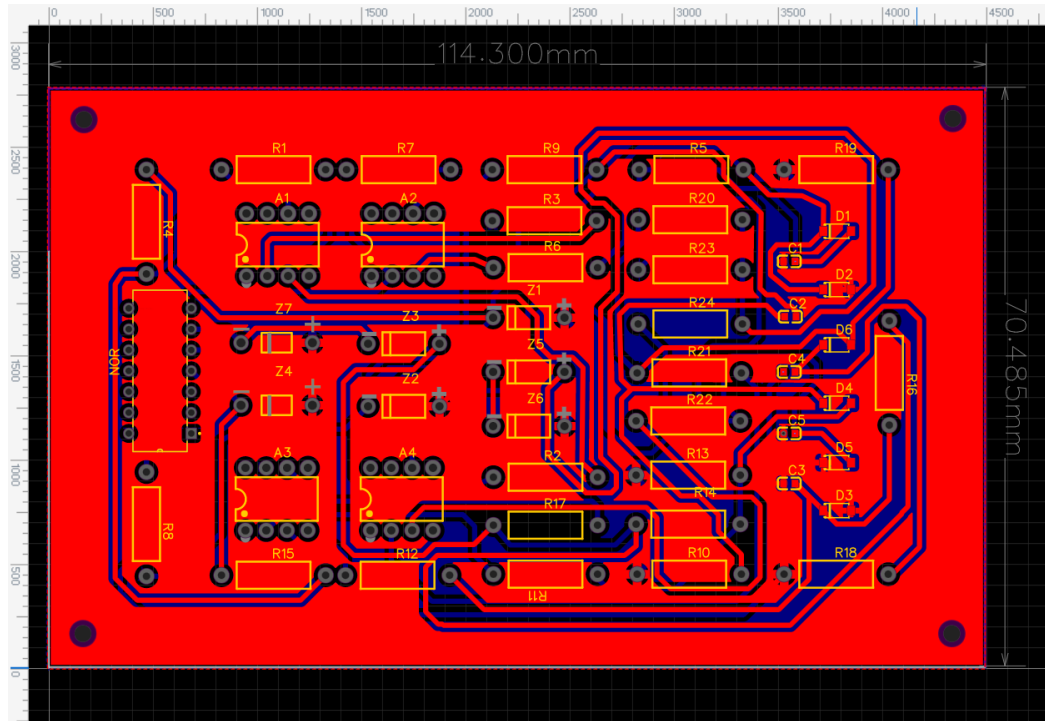
แบบ Through-Hole Technology (THT)



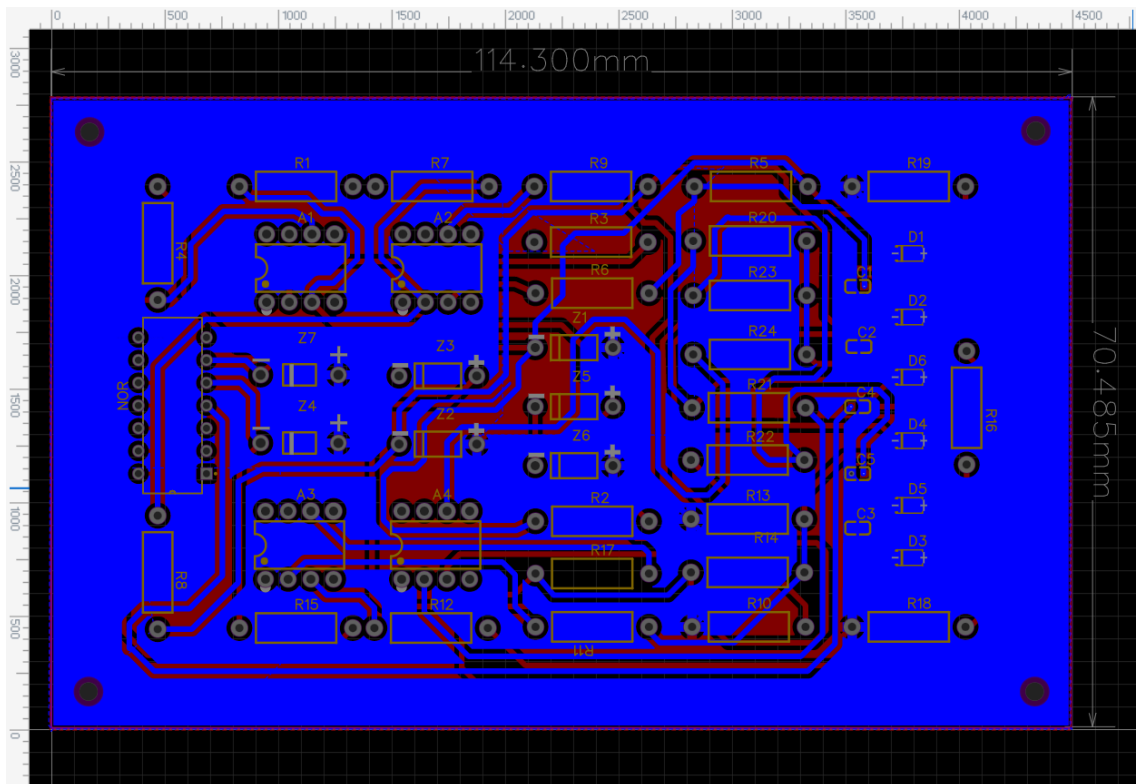
Printed circuit board Routing 2 Layers for Through-Hole Technology (THT)



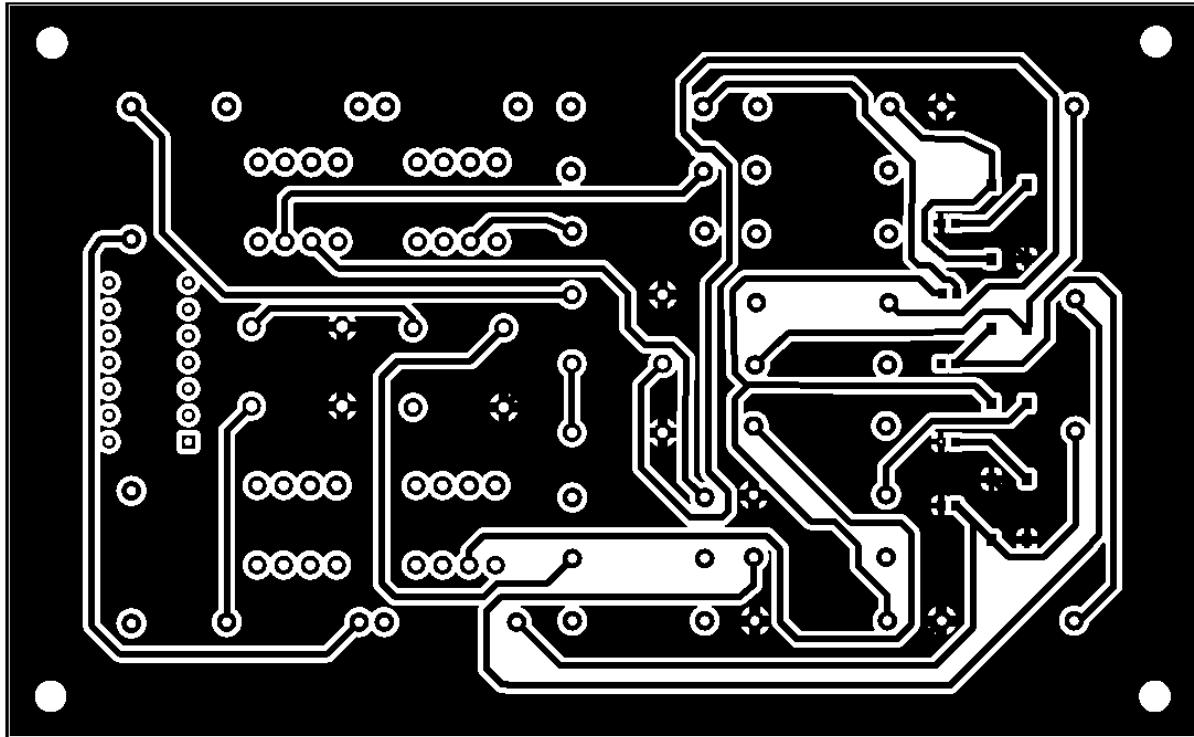
Route the PCB ground on the top layer for Through-Hole Technology (THT)



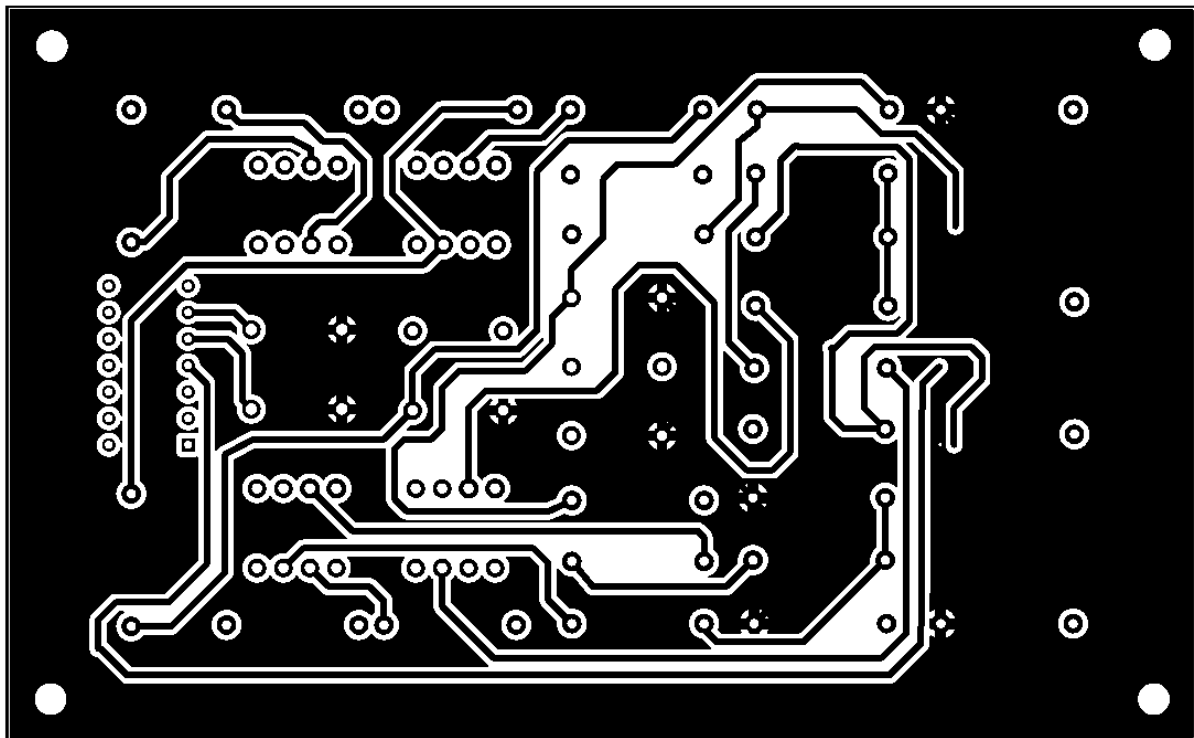
Route the PCB ground on the bottom layer for Through-Hole Technology (THT)



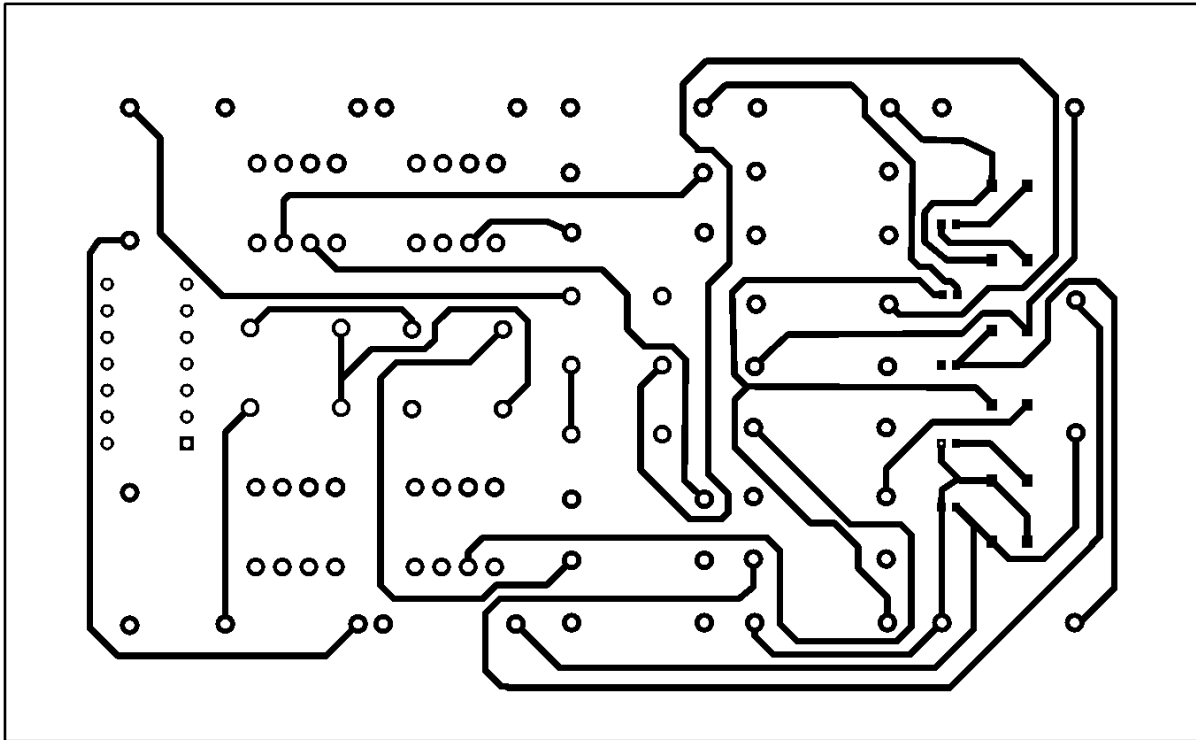
PCB Print trace line and ground on the top layer for Through-Hole Technology (THT)



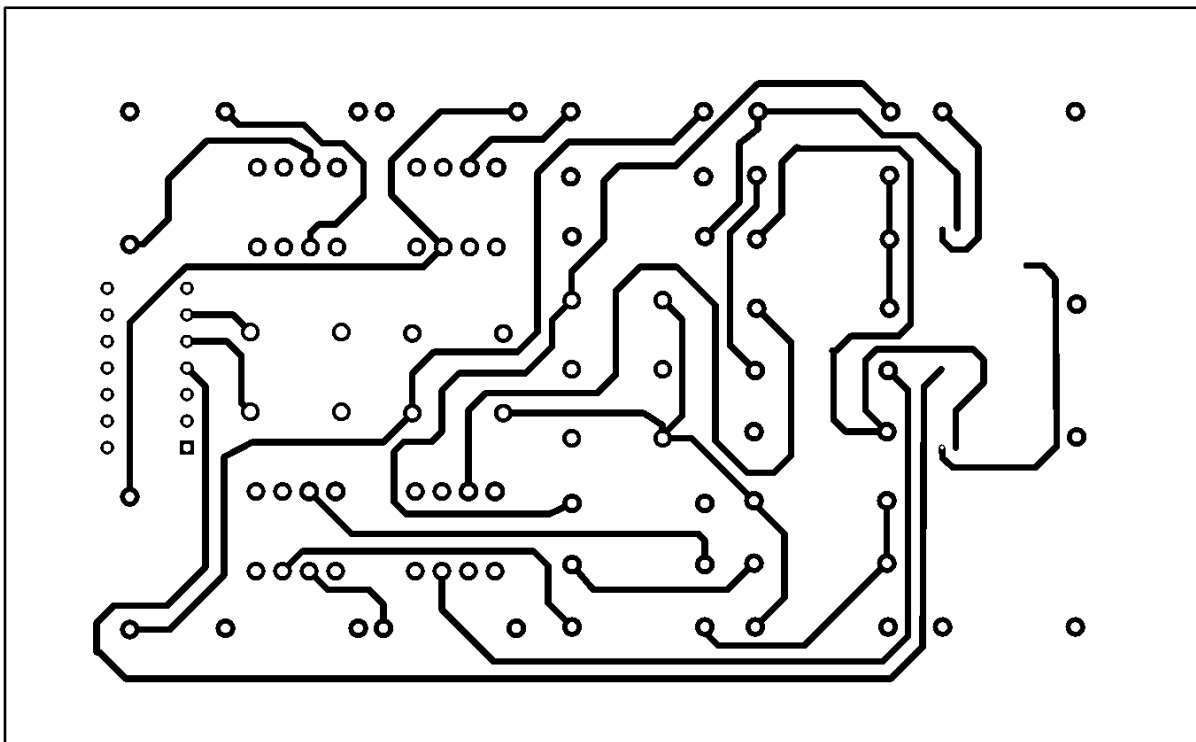
PCB Print trace line and ground on the bottom layer for Through-Hole Technology (THT)



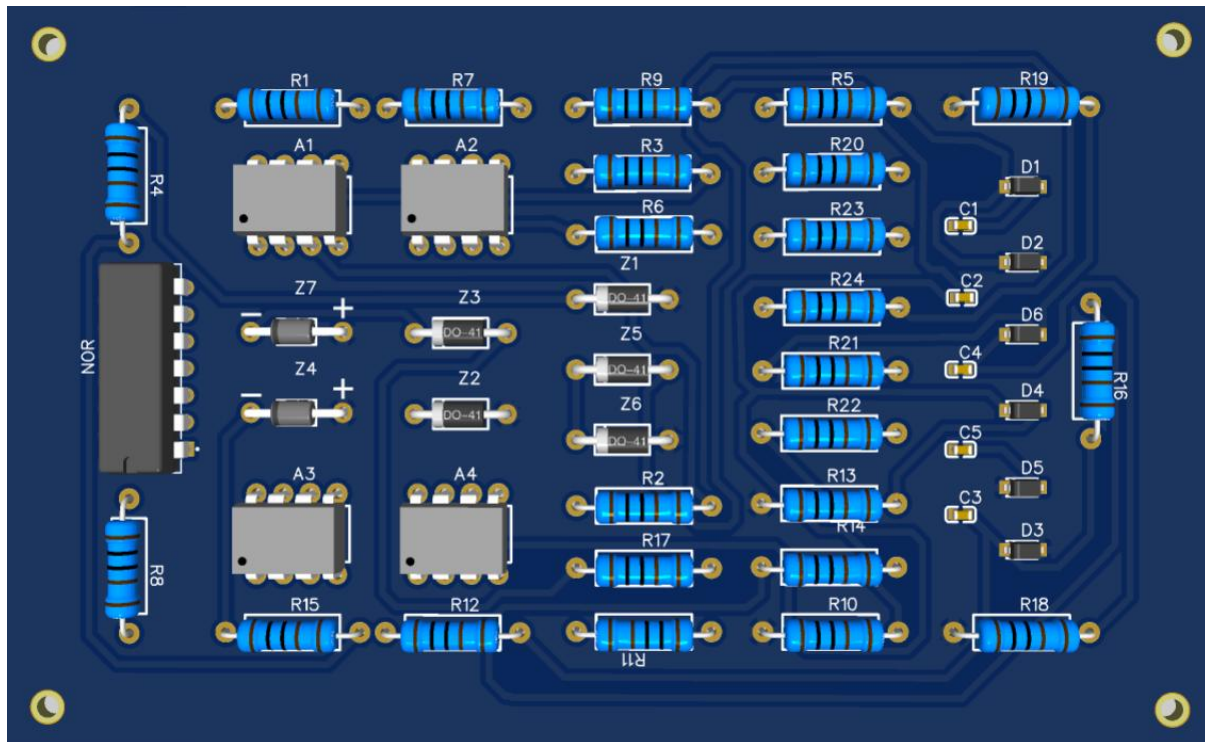
PCB Print trace line on the top layer for Through-Hole Technology (THT)



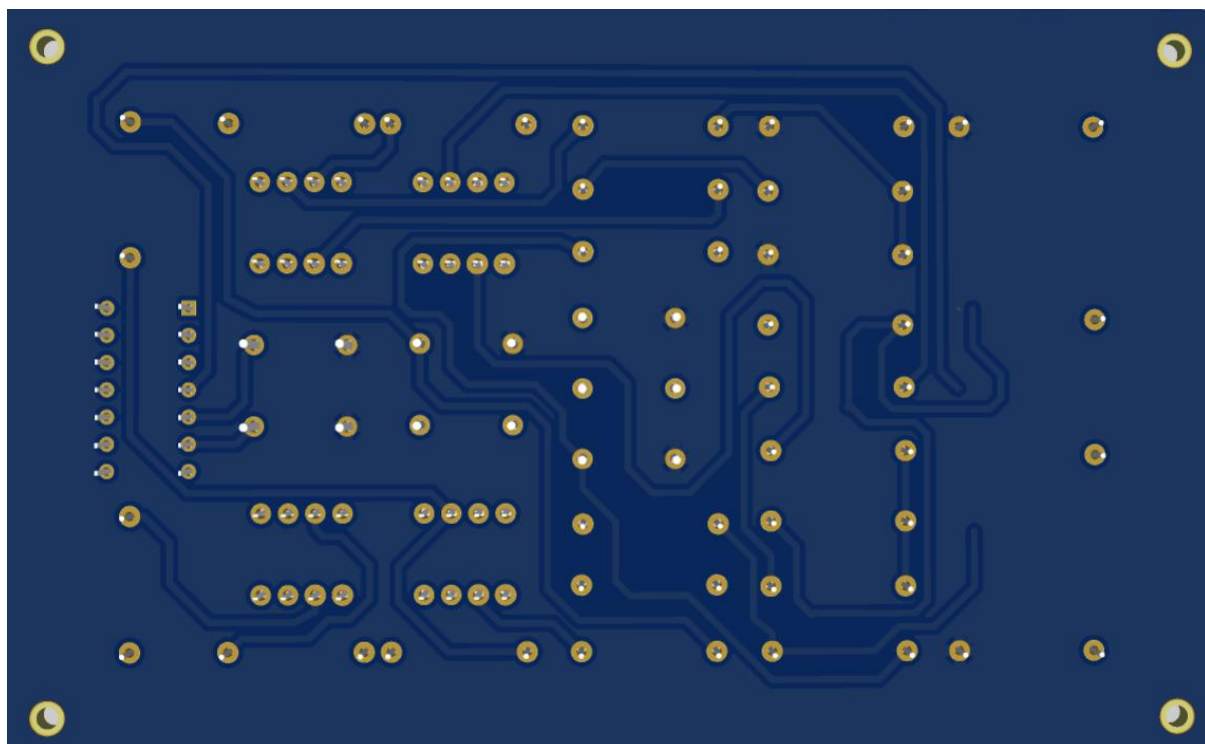
PCB Print trace line on the bottom layer for Through-Hole Technology (THT)



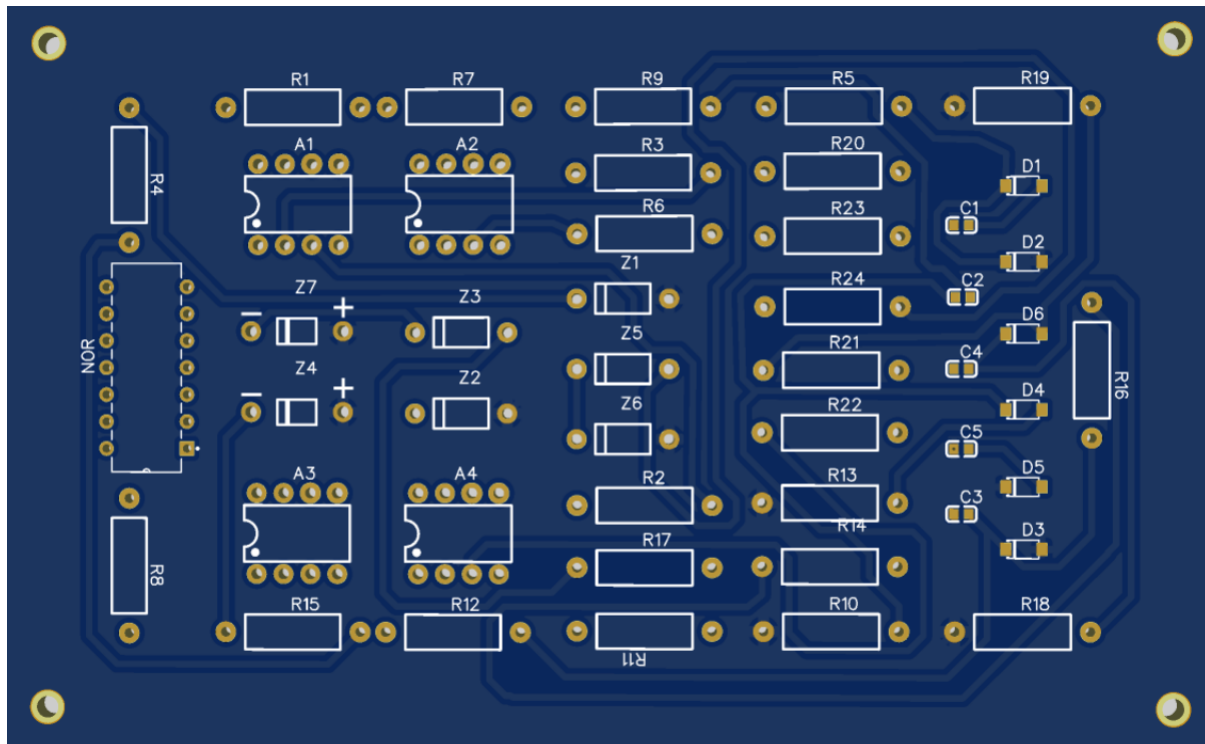
The top layer of the 3D model is visible for Through-Hole Technology (THT)



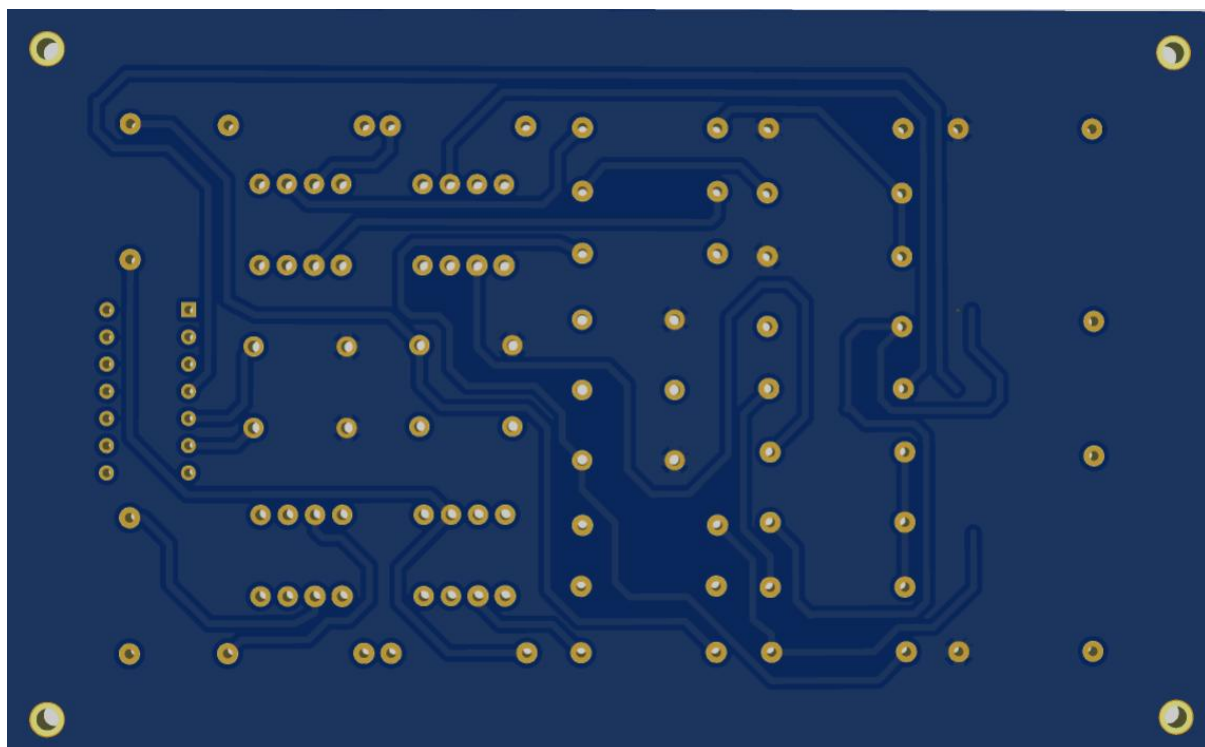
The bottom layer of the 3D model is visible for Through-Hole Technology (THT)



The top layer of the 3D model is invisible for Through-Hole Technology (THT)



The bottom layer of the 3D model is invisible for Through-Hole Technology (THT)



ข้อมูลของ PCB แบบ Through-Hole Technology (THT)

ขนาดแผ่น 114.300 × 70.485 mm.

ชื่ออุปกรณ์		ขนาด (mm)
Nor-gate_7402		19.558 × 6.604 mm.
1N4148WT		3.000 × 1.700 mm.
Zener Diode	1N4566	3.600 × 3.000 mm.
	1N752A	5.000 × 3.000 mm.
LM741_Op-Amp		9.906 × 5.207 mm.
Resistor		9.000 × 3.300 mm.
Capacitor		3.000 × 1.700 mm.

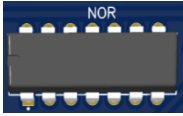
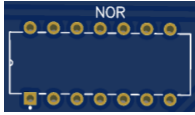
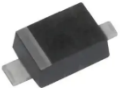
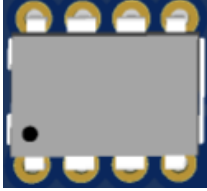
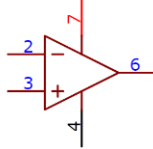
สายไฟหนา 0.635 mm

ระยะห่างขั้นต่ำของสายไฟ 0.508 mm

รูสำหรับเจาะเพื่อใส่อุปกรณ์เส้นผ่านศูนย์กลาง 1.000 mm

ทองแดงรอบรูสำหรับเจาะเส้นผ่านศูนย์กลาง 1.800 mm

ชื่ออุปกรณ์และสัญลักษณ์ของอุปกรณ์ที่ใช้แบบ Surface Mount Technology (SMT)

ชื่ออุปกรณ์		ภาพอุปกรณ์	3D PCB	สัญลักษณ์	สัญลักษณ์บน PCB
Nor-gate_7402					
1N4148WT_SMD					
Zener Diode	1N4566 SMD				
	1N752A SMD				
LM741_Op-Amp					
Resistor_SMD					
Capacitor_SMD					

หมายเหตุ

1N4148WT_SMD : D₁, D₂, D₃, D₄, D₅, D₆

1N4566 : Z₁, Z₂, Z₃, Z₅, Z₆

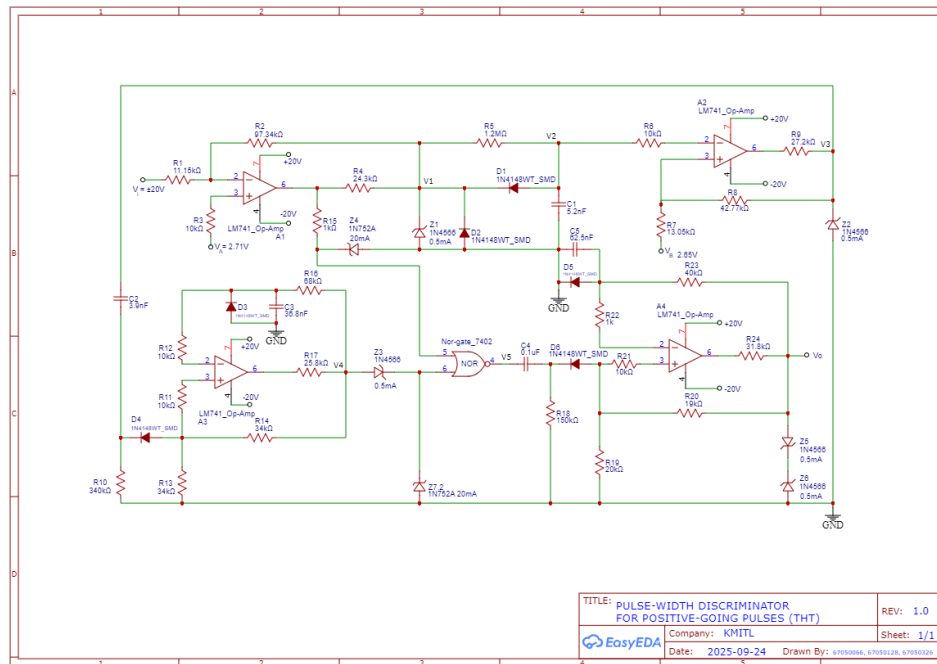
1N752A : Z₄, Z₇

LM741_Op-Amp : A₁, A₂, A₃, A₅

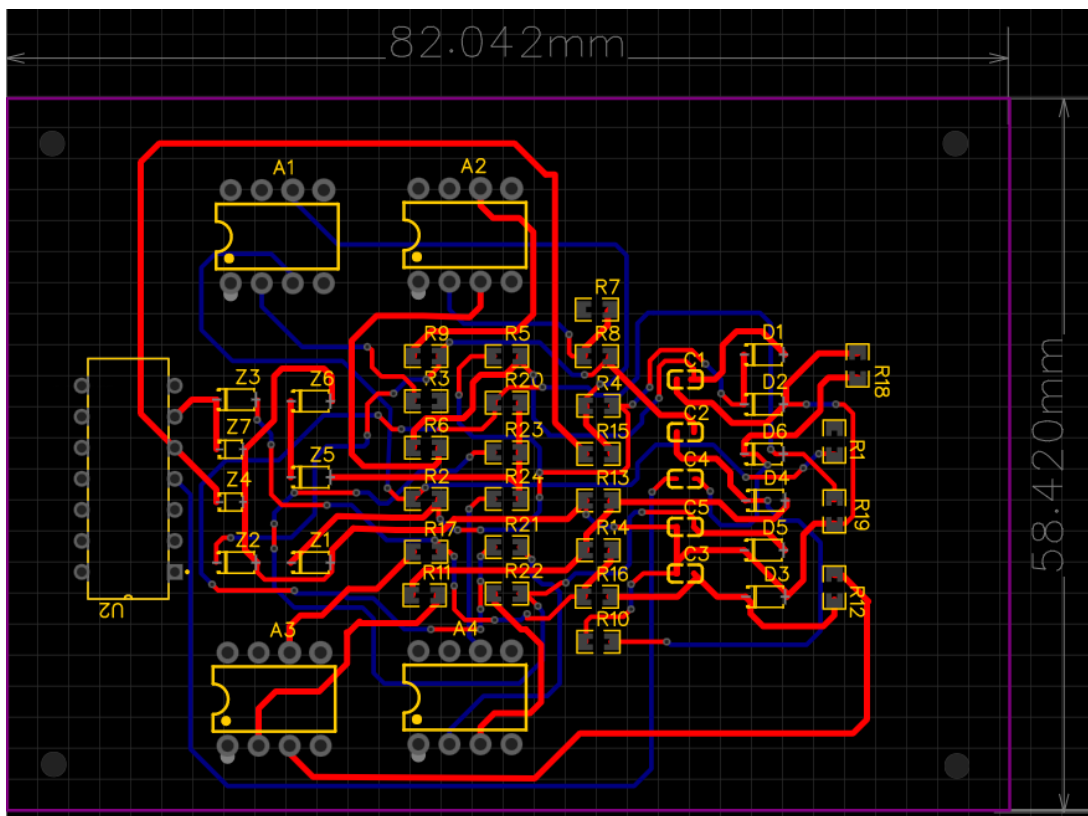
Surface Mount Technology (SMT) : แบบบัดกรีติดบน Top layer

Schematic Diagram of PULSE-WIDTH DISCRIMINATOR FOR POSITIVE-GOING PULSES

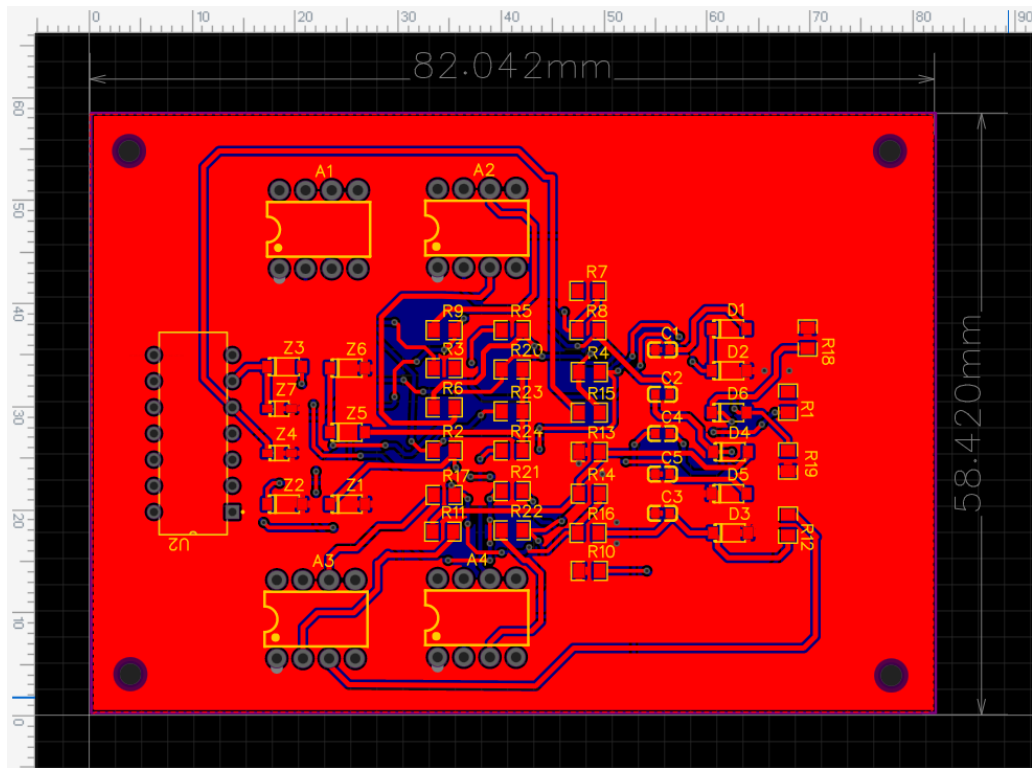
แบบ Surface Mount Technology (SMT)



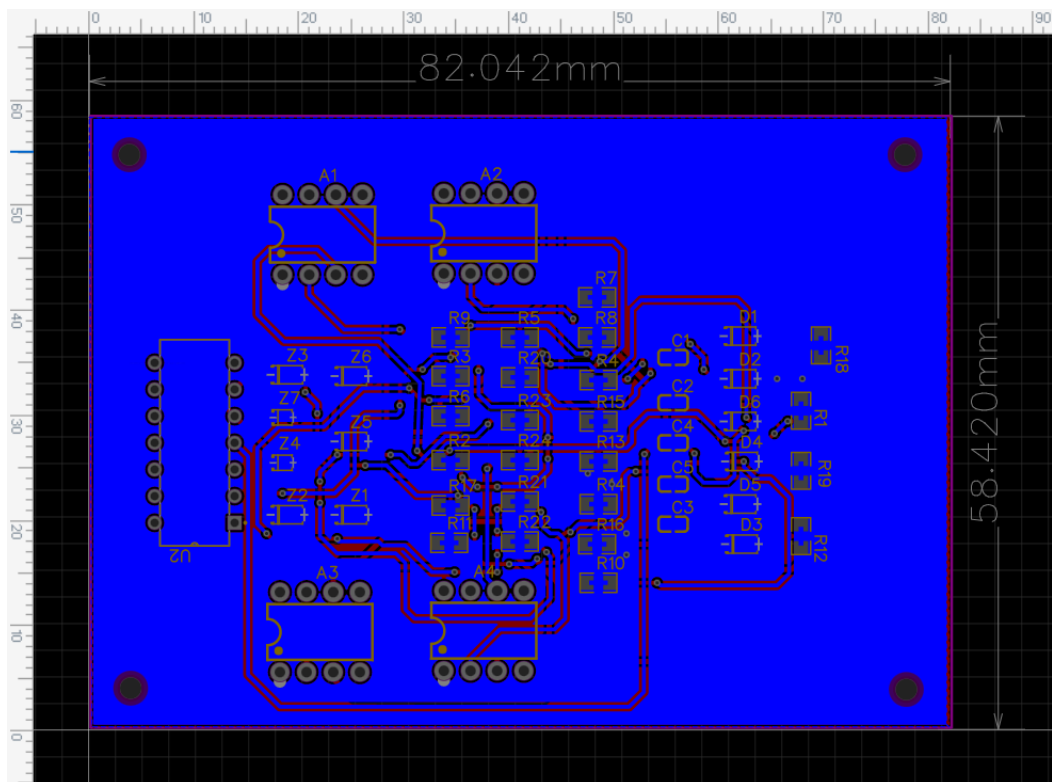
Printed circuit board Routing 2 Layers for Surface Mount Technology (SMT)



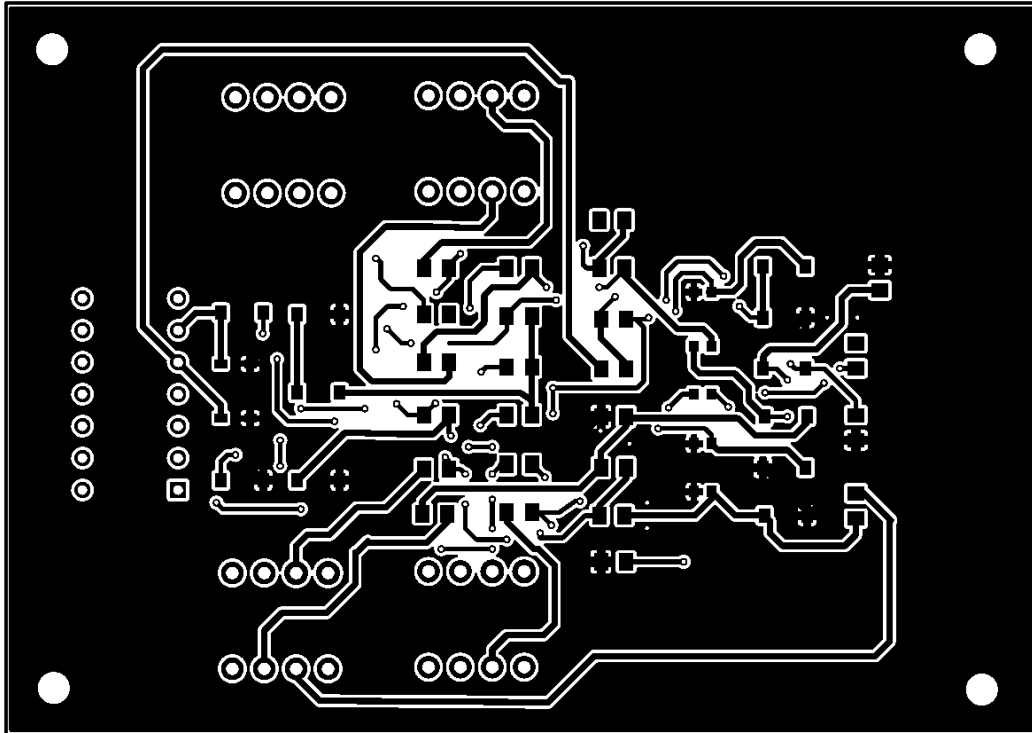
Route the PCB ground on the top layer for Surface Mount Technology (SMT)



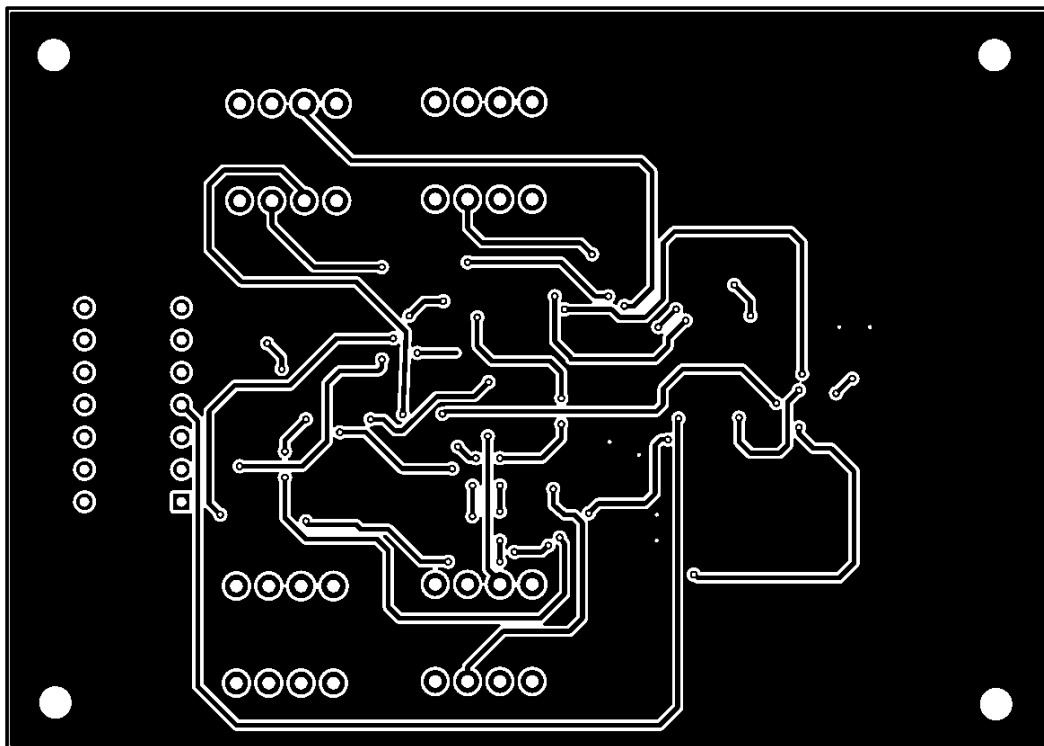
Route the PCB ground on the bottom layer Surface Mount Technology (SMT)



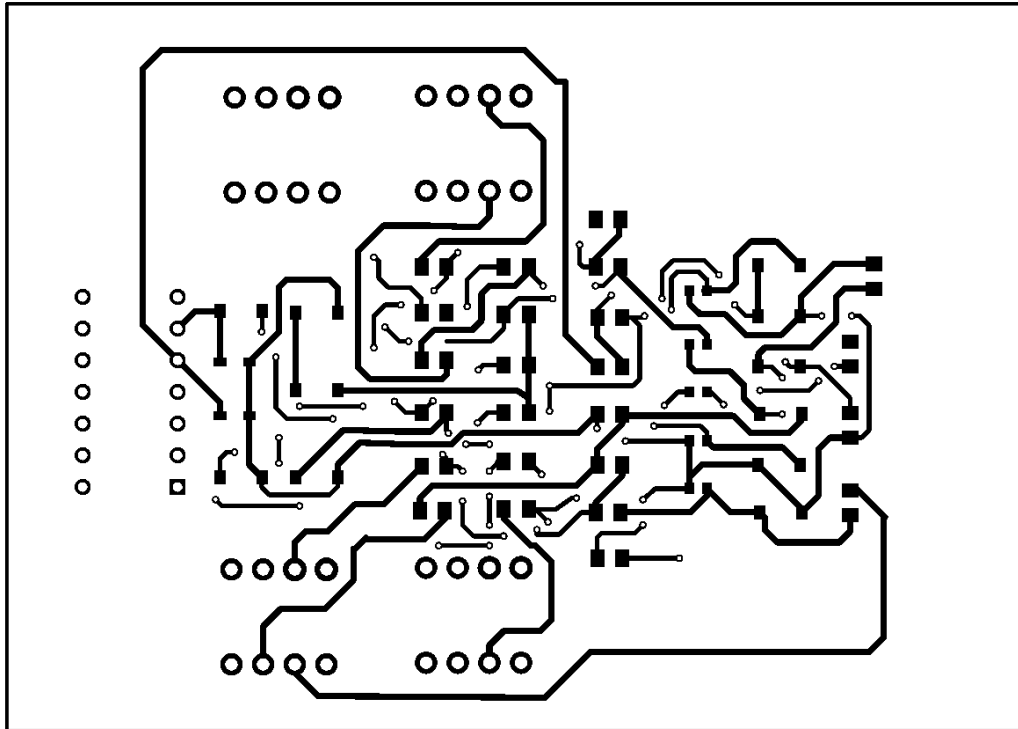
PCB Print trace line and ground on the top layer for Surface Mount Technology (SMT)



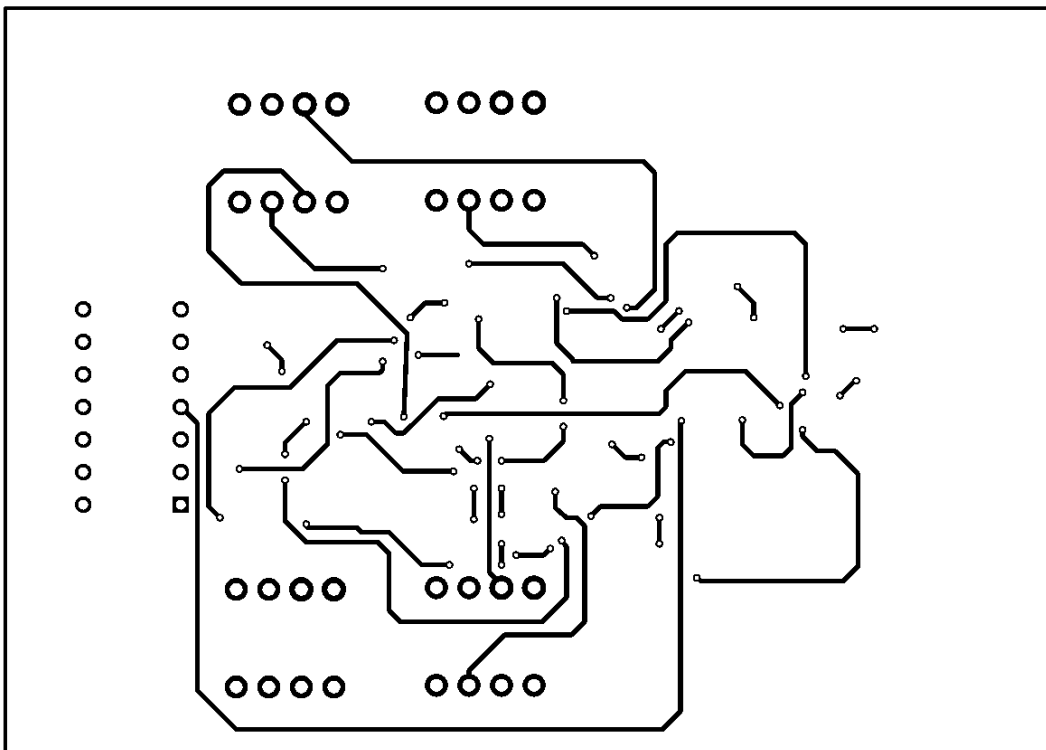
PCB Print trace line and ground on the bottom layer for Surface Mount Technology (SMT)



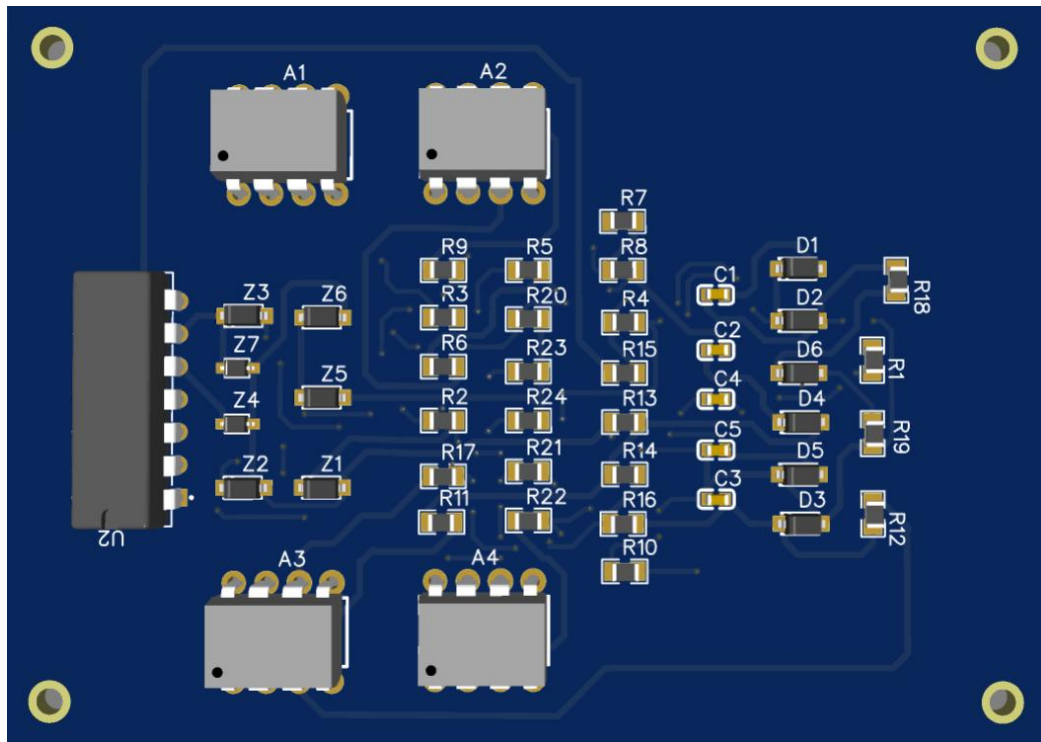
PCB Print trace line on the top layer for Surface Mount Technology (SMT)



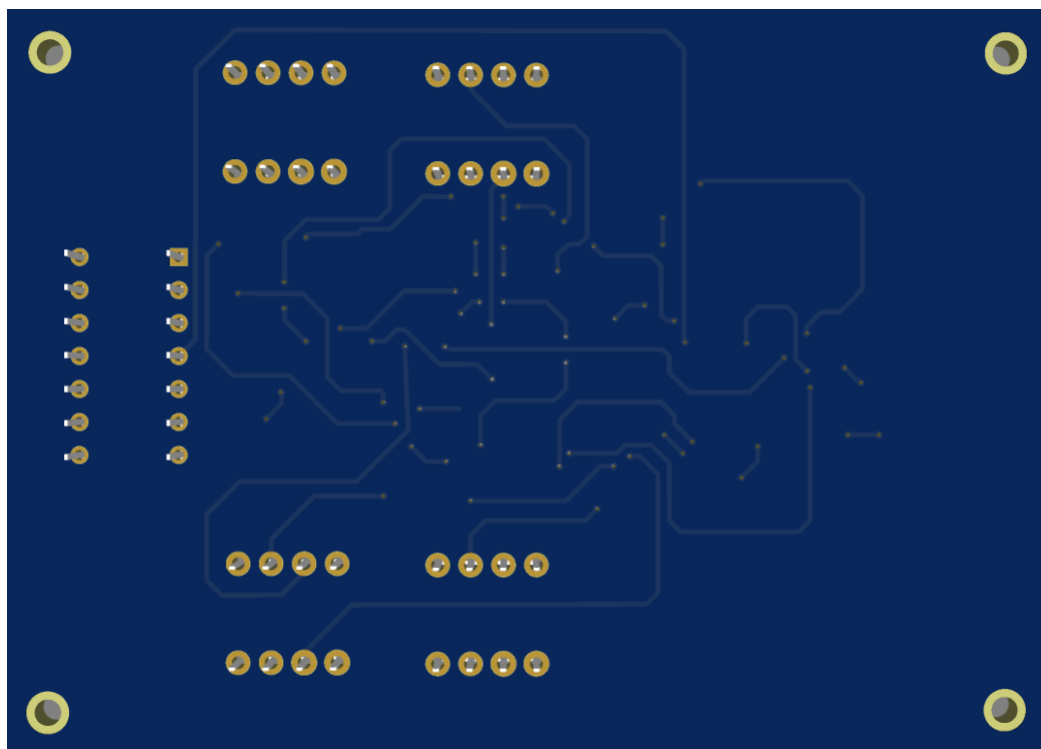
PCB Print trace line on the bottom layer for Surface Mount Technology (SMT)



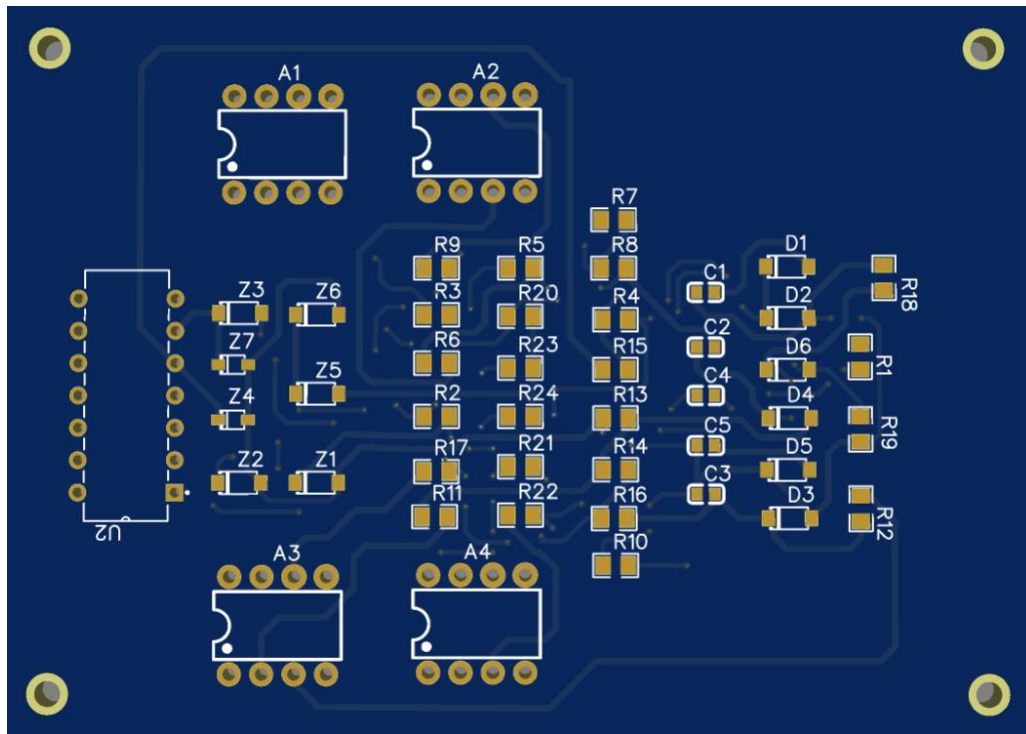
The top layer of the 3D model is visible for Surface Mount Technology (SMT)



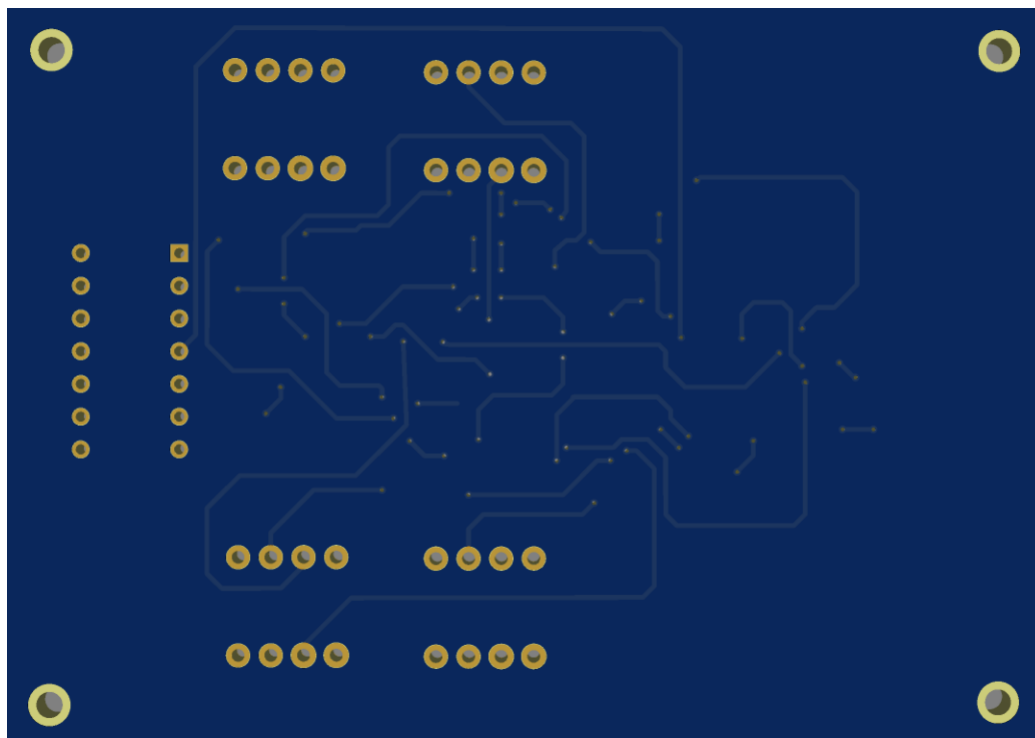
The bottom layer of the 3D model is visible for Surface Mount Technology (SMT)



The top layer of the 3D model is invisible for Surface Mount Technology (SMT)



The bottom layer of the 3D model is invisible for Surface Mount Technology (SMT)



ข้อมูลของ PCB แบบ Surface Mount Technology (SMT)

ขนาดแผ่น 82.042 × 58.420 mm.

ชื่ออุปกรณ์		ขนาด (mm)
Nor-gate_7402		19.558 × 6.604 mm.
1N4148WT		3.000 × 1.700 mm.
Zener Diode	1N4566	3.000 × 2.000 mm.
	1N752A	3.000 × 2.000 mm.
LM741_Op-Amp		9.906 × 5.207 mm.
Resistor		3.500 × 1.800 mm.
Capacitor		3.000 × 1.700 mm.

สายไฟหนา 0.635 mm

ระยะห่างขั้นต่ำของสายไฟ 0.508 mm

รูสำหรับเจาะเพื่อใส่อุปกรณ์เส้นผ่านศูนย์กลาง 1.000 mm

ทองแดงรอบรูสำหรับเจาะเส้นผ่านศูนย์กลาง 1.800 mm

Design requirements

$$T_p = 10 \text{ ms with } +5V \text{ amplitude and } 10 \text{ ms spacing between pulses}$$

$$T_t = 20\% \text{ of } T_p \text{ or } 2 \text{ ms}$$

$$V_i = \pm 5V \text{ composite waveform with } 1 \text{ V noise (peak-to-peak)}$$

$$V^+ = 20 \text{ V}, \quad V^- = -20 \text{ V}$$

$$T_o = 2 \text{ ms} + 7.5V \text{ standby}, \quad -7.5V \text{ during pulse}$$

Device data

$$\frac{\Delta V_{io1}}{\Delta T} = \frac{\Delta V_{io2}}{\Delta T} = 5 \text{ mV } (-55 \text{ to } +125^\circ\text{C})$$

$$\frac{\Delta I_{b1}}{\Delta T} = \frac{\Delta I_{b2}}{\Delta T} = 500 \text{ nA } (-55 \text{ to } +125^\circ\text{C})$$

$$I_{z1} = I_{z2} = I_{z3} = I_{z5} = I_{z6} = 0.5 \text{ mA (1N4566, 6.4 V)}$$

$$I_{z4} = 20 \text{ mA (1N752A, 6.8 V)}$$

$$S_3 = S_4 = 0.5 \text{ V}/\mu\text{s}$$

Step 1 Equation 9 produces

$$R_3 = \frac{\Delta V_{io1}/\Delta T}{\Delta I_{b1}/\Delta T} = \frac{5 \text{ mV}}{50000 \text{ mA}} = 10 \text{ k}\Omega$$

Step 2 A IN4566 reference diode (6.4 V) is chosen for V_{Z1} . Its temperature coefficient is 0.005%/°C so it will contribute an error to T_p which can be calculated using Eq.1.

Step 3 The trip level V_{L1} of the first level detector is chosen to be 3 V. This level is halfway between the peak noise and the peak voltage of the input pulses T_p . The hysteresis width ΔV_1 is chosen to be 1 percent of 3 V, or 30 mV.

R_1 is now computed:

$$R_1 = \frac{(V_{Z1} + V_{D1})R_3}{V_{Z1} - \Delta V} = \frac{(6.4V + 0.7V)10k\Omega}{6.4V - 0.03V} = 11.15 \text{ k}\Omega$$

Step 4. Equation 11 now provides us with R_2

$$R_2 = \frac{R_1 R_3}{R_1 - R_3} = \frac{(11.145 \text{ k}\Omega)(10 \text{ k}\Omega)}{11.145 \text{ k}\Omega - 10 \text{ k}\Omega} = 97.34 \text{ k}\Omega$$

Step 5. The bias voltage V_A is now computed:

$$V_A = \frac{R_2 V_{L1} - R_1 V_{D1}}{R_1 + R_2} = \frac{(97.336k\Omega)(3V) - (11.145k\Omega)(0.7V)}{11.145k\Omega + 97.336k\Omega} = 2.62 \text{ V}$$

Step 6. The required R_4 is

$$R_4 = \frac{V^+ - V_{Z1}}{I_{Z1}} = \frac{20V - 6.4V}{0.5 \text{ mA}} = 27.2 \text{ k}\Omega$$

Therefore, R_4 is 17 times less than $R_2 = 78.96k\Omega$. This will make I_z lower than the required 0.5 mA by 0.047 mA. If R_4 is lowered to

$$R_4 = \frac{V^+ - V_{Z1}}{I_{Z1} + (V_{Z1}/(R_1 + R_2))} = \frac{20V - 6.4V}{0.5 \text{ mA} + (6.4V/(11.145k\Omega + 97.336k\Omega))} = 24.33k\Omega$$

Then I_{Z1} will remain at 0.5 mA

Step 7. The required R_6 is now found from Eq.10

$$R_6 = \frac{\Delta V_{io2}/\Delta T}{\Delta I_{b2}/\Delta T} = \frac{5 \text{ mV}}{50000 \text{ mA}} = 10 \text{ k}\Omega$$

Step 8. R_9 is calculated as follows:

$$R_9 = \frac{V^+ - V_{Z2}}{I_{Z2}} = \frac{20V - 6.4V}{0.5 \text{ mA}} = 27.2 \text{ k}\Omega$$

Step 9. Let $V_{L3} = 3 \text{ V}$. This is halfway up the $R_5 C_1$ charging curve which levels off at $V_{1max} = 6.4 \text{ V}$. If V_{L2} is more than 80 percent V_{1max} , small variations in R_5 and C_1 will begin to have an appreciable effect on stability.

Let. $\Delta V_{Z2} = 0.1 V_{L2} = 0.1(3V) = 300 \text{ mV}$. R_7 is now found from

$$R_7 = \frac{(V_{Z2} + V_{DZ2})R_6}{V_{Z2} + V_{DZ2} - \Delta V} = \frac{(6.4V + 0.7V)10\text{k}\Omega}{6.4V + 0.7V - 0.3V} = 10.44 \text{ k}\Omega$$

Step 10. Equation 12 now provides us with R_8

$$R_8 = \frac{R_6 R_7}{R_7 - R_6} = \frac{(10.44 \text{ k}\Omega)(10 \text{ k}\Omega)}{10.44 \text{ k}\Omega - 10 \text{ k}\Omega} = 237.27 \text{ k}\Omega$$

Step 11. The bias voltage V_B for the second level detector is now computed:

$$V_B = \frac{V_{L2}(R_7 + R_8) - V_{Z2}R_7}{R_8} = \frac{(3V)(10.44\text{k}\Omega + 237.27\text{k}\Omega) - (10.44\text{k}\Omega)(0.7V)}{237.27\text{k}\Omega} = 3.1 \text{ V}$$

Step 12. The $R_5 C_1$ product is computed from Eq.1:

$$R_5 C_1 = \frac{T_p - \Delta T_p}{\ln[V_{Z1}/(V_{Z1} - V_{L2})]} = \frac{(0.01 - 0.001)\text{ms}}{\ln[6.4V/(6.4V - 3V)]} = 0.0142$$

As a first approach, let

$$R_5 = 100R_4 = 2720 \text{ k}\Omega$$

Then

$$C_1 = \frac{0.0142\text{ms}}{2720 \text{ k}\Omega} = 5.22 \text{ pF}$$

Step 13. The Z_4 is chosen to be compatible with TTL logic. The calculated R_{17} is

$$R_{17} = \frac{V^+ - V_{Z4}}{I_{Z4}} = \frac{20V - 6.8V}{20 \text{ mA}} = 660 \Omega$$

Step 14. $R_{13} - R_{14}$ voltage divider is calculated from

$$R_{13} = R_{14} = \frac{100V_{Z4}}{I_{Z4}} = \frac{100(6.8V)}{20 \text{ mA}} = 34 \text{ k}\Omega$$

Step 15. Assuming A_3 is a 741 type of op amp, we can let

$$R_{11} = R_{12} = 10 \text{ k}\Omega$$

We also compute

$$R_{10} = 10R_{13} = 340 \text{ k}\Omega$$

Step 16. The C_2 is calculated from

$$C_2 = \frac{T_t}{R_{10}} = \frac{2 \text{ ms}}{340 \text{ k}\Omega} = 5.88 \text{ nF}$$

Step 17. R_{16} and C_3 are determined.

$$R_{16} = 2R_{13} = 68 \text{ k}\Omega$$

$$C_3 = \frac{T_t}{R_{16}} = \frac{2 \text{ ms}}{(0.8)68 \text{ k}\Omega} = 36.76 \text{ nF}$$

Step 18. The calculated R_{17} is

$$R_{17} = \frac{V^+ - V_{Z6}}{I_{Z6}} = \frac{20 \text{ V} - (6.4 + 0.7) \text{ V}}{0.5 \text{ mA}} = 25.8 \text{ k}\Omega$$

Step 19. We next compute

$$R_{19} = \frac{50\Delta v_5}{I_{Z6}} = \frac{50(2 \text{ V})}{0.5 \text{ mA}} = 20 \text{ k}\Omega$$

$$R_{20} = \frac{100V_{Z6}}{I_{Z6}} - R_{19} = \frac{100(6.4 \text{ V})}{0.5 \text{ mA}} - 20 \text{ k}\Omega = 19 \text{ k}\Omega$$

Step 20. Assuming A_4 is a 741 type of op amp, we can let

$$R_{21} = R_{22} = 10 \text{ k}\Omega$$

We next compute

$$R_{18} = 10R_{19} = 10(20 \text{ k}\Omega) = 200 \text{ k}\Omega$$

Step 21. The C_4 is calculated from

$$C_4 = \frac{T_o}{R_{18}} = \frac{2 \text{ ms}}{200 \text{ k}\Omega} = 10 \text{ nF}$$

Step 24. R_{23} and C_5 are the last values to be determined.

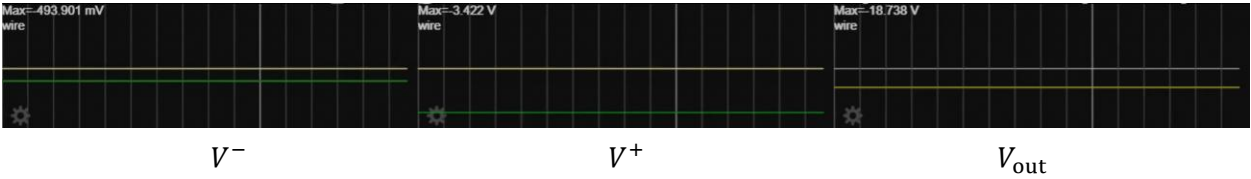
$$R_{23} = 2R_{19} = 2 \cdot 20 \text{ k}\Omega = 40 \text{ k}\Omega$$

$$C_5 = \frac{T_o}{0.8R_{23}} = \frac{2 \text{ ms}}{0.8(40 \text{ k}\Omega)} = 62.5 \text{ nF}$$

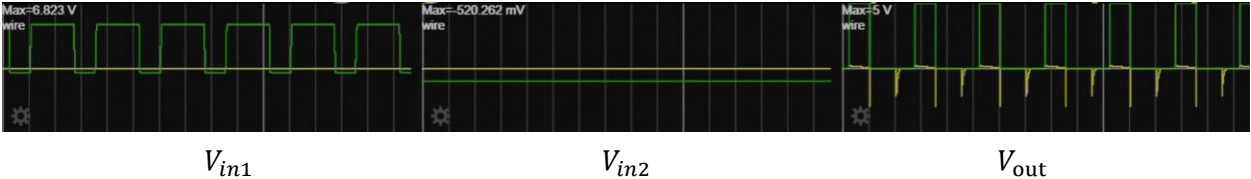
Three oscilloscope waveforms are shown side-by-side. The first waveform, labeled V^- , is a sine wave with a peak value of 4.425 V. The second waveform, labeled V^+ , is a flat line at 2.62 V. The third waveform, labeled V_{out} , is a square wave with a peak value of 18.999 V.

Three oscilloscope waveforms are displayed side-by-side. The first waveform, labeled V^- , shows a periodic sawtooth signal with a maximum value of 2.374 V. The second waveform, labeled V^+ , shows a constant high-level signal with a maximum value of 3.266 V. The third waveform, labeled V_{out} , shows a constant high-level signal with a maximum value of 19.394 V.

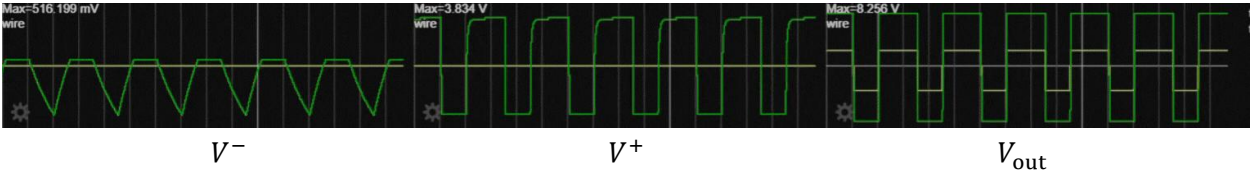
Waveform ที่ V_{A3}



Waveform ที่ V_{NOR}



Waveform ที่ V_{A4}



โปรแกรม :

การวาดลายวงจร และ การเขียนแบบวงจร : EasyEDA 6.5.51

การ simulate วงจร : <https://www.falstad.com/circuit/circuitjs.html>

ลิ้งค์วงจร : <https://tinyurl.com/28m83jwr>